

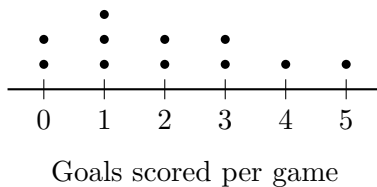
Measuring the Centre and Spread of a Data Set

Finding the mode, median and mean of a data set, measuring its spread with the range and the interquartile range, and deciding which of those numbers actually describes the group

How to use this sheet.

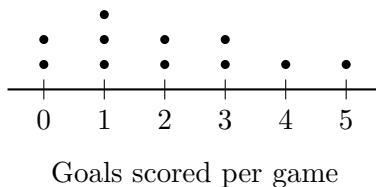
- A *measure of centre* answers “what is a typical value here?” — mode, median and mean are three different answers to that one question.
- A *measure of spread* answers “how much do the values differ from each other?” — the range and the interquartile range are two different answers to that one.
- Put the data in order before you touch the median, the quartiles or the range. On a dot plot that ordering is already done for you: read the dots from left to right.
- On a dot plot the label on the axis is a *value* and each dot is one *item*. Two dots above the 4 means two items worth 4 each.

1. The mode. The dot plot shows how many goals the Year 7 team scored in each of its eleven games. Each dot is one game.



Which number of goals came up most often? That value is the *mode*.

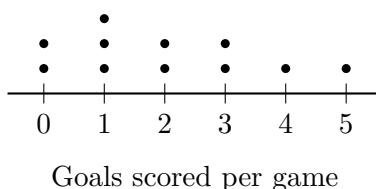
Answer: _____

2. The median. Same eleven games.

Reading the dots from left to right already puts the eleven values in order, so the median is the value the middle dot sits on. Count in from each end until you meet.

Which position is the middle one out of eleven? _____

Answer: _____

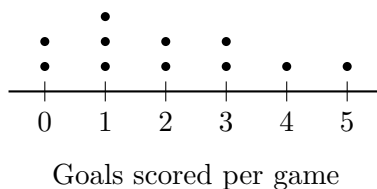
3. The mean. Same eleven games.

Add the goals from all eleven games, then divide by the number of games. Two dots above the 4 would mean two games of four goals each — count the dots, not the labels.

Total goals: _____ Number of games: _____

Answer: _____

4. The range. Same eleven games.

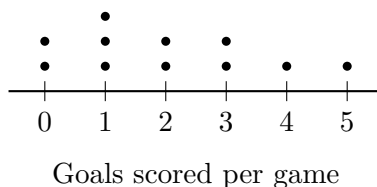


The range measures spread: how far apart the two most extreme games were.

Most goals: _____ Fewest goals: _____

Answer: _____

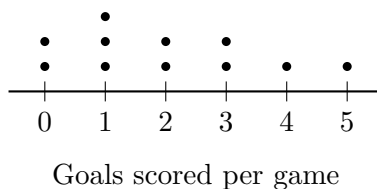
5. Quartiles. Same eleven games. Split the ordered list at the median, leaving that middle value out of both halves.



The lower quartile Q_1 is the median of the lower half; the upper quartile Q_3 is the median of the upper half. Write both halves out before you start.

$Q_1 =$ _____ $Q_3 =$ _____

6. The interquartile range. Same eleven games.

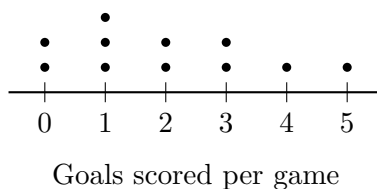


The interquartile range is $Q_3 - Q_1$: the width of the middle half of the data.

Underneath your working, write one sentence saying what that number tells you about a typical game.

Answer: _____

7. One game at random. Same eleven games. One of them is picked at random, and every game is equally likely.



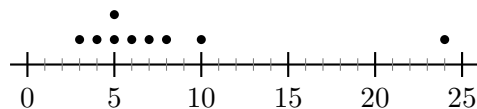
Count the *games* that finished with 3 or more goals — not the labels on the axis. Give the probability as a fraction.

Favourable games: _____ Games in total: _____

Answer: _____

8. A different group. Nine students recorded how many books they read over the summer. Here are their totals, already in order, with the same data as a dot plot underneath.

3, 4, 5, 5, 6, 7, 8, 10, 24



Books read over the summer

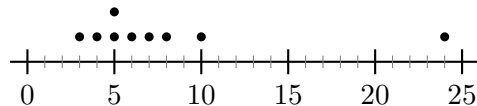
Find the median number of books.

Which position is the middle one out of nine? _____

Answer: _____

9. The mean of the same nine.

3, 4, 5, 5, 6, 7, 8, 10, 24



Books read over the summer

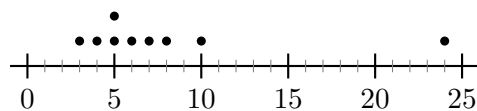
Add all nine totals, then divide by nine.

Sum of the nine totals: _____

Answer: _____

10. The range of the same nine.

3, 4, 5, 5, 6, 7, 8, 10, 24



Books read over the summer

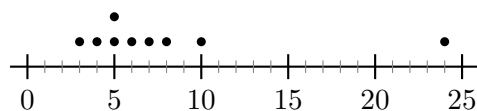
Largest: _____ Smallest: _____

Which single student decides this answer? _____

Answer: _____

11. Devi is the student who read 24. These nine students have a mean of 8 books and a median of 6 books — you worked both out above.

3, 4, 5, 5, 6, 7, 8, 10, 24



Books read over the summer

(a) Work out the mean of the other eight students, leaving Devi out.

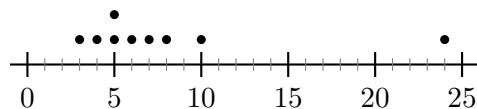
Sum without Devi: _____ Mean of the eight: _____

(b) What happens to the median when Devi is left out? _____

(c) Which measure of centre — mean or median — describes this group of readers better? Write two or three sentences, and say what Devi's total does to each one.

12. Two ways to measure spread. The same nine students, and the range you found above.

3, 4, 5, 5, 6, 7, 8, 10, 24



Books read over the summer

(a) Find Q_1 and Q_3 for these nine values, then the interquartile range.

$Q_1 =$ _____ $Q_3 =$ _____ IQR: _____

(b) Which is larger, the range or the IQR? _____

(c) Which of the two gives a fairer picture of how spread out this group is? Write two or three sentences saying what each one actually measures.
